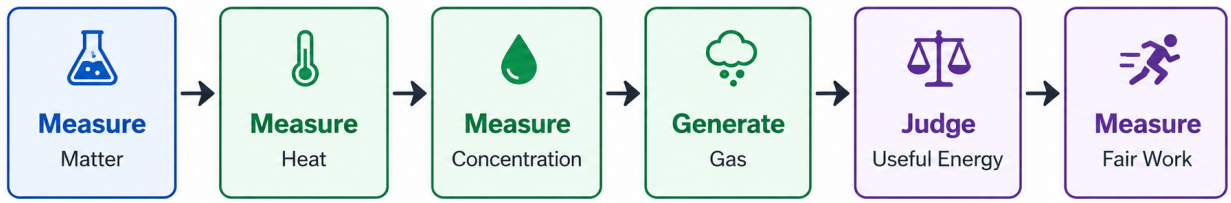
UMA Learning Sequence

Phenomena-First Experimental Order and Precedence

Corrected professional summary for curriculum development
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This document captures the corrected ranking and order of execution for the UMA learning sequence using a **phenomena-first** criterion. The emphasis is no longer on which sensor, microcontroller, or dashboard is available first. Instead, the sequence begins with the simplest visible phenomenon, adds trustworthy measurement, then adds modeling, and only then asks whether the result is useful, efficient, fair, or enough.

Core principle: The technology appears when the phenomenon demands a better witness. The scale witnesses mass loss. The thermometer witnesses heat flow. The light measurement witnesses concentration. The balloon witnesses gas. The dashboard witnesses change over time. Fair Fitness witnesses human work without humiliation.



Central thread: measure matter → measure heat → measure concentration → generate gas → judge useful energy → measure fair work

A. Executive Summary

The corrected learning progression is: **measure matter -> measure time/rate -> measure temperature -> measure heat transfer -> measure concentration -> generate gas -> measure fuel-to-heat conversion -> design human fuel -> measure human work -> judge fairness.**

The sequence is ranked by meaningful precedence criteria: (1) What phenomenon is easiest to see? (2) What variable must be understood first? (3) What measurement unlocks the next lab? (4) What model becomes meaningful only after prior experience? (5) Where does the work become DYNAMETICS - useful energy, cost, fuel, work, fairness?

Criterion	Operational meaning
Phenomenon first	Begin with the simplest visible change before adding hardware complexity.
Measurement second	Establish trusted ways to observe the phenomenon.
Modeling third	Add graphs, curves, stoichiometry, and efficiency only after the raw process is familiar.
Judgment last	Use DYNAMETICS and Fair Fitness only after students can measure what changed and why.

B. Correct Placement Rules

Item	Correct placement	Reason
Purple Power	Tier 3 only	A DYNAMETICS / Bio-DYNAMETICS fuel activity, not a Kitchen Lab activity.
CandleLab / PopCan Calorimeter	Tier 3 only	A useful-energy / efficiency synthesis, not merely a Kitchen Lab thermodynamics lab.
TeaTime / Beer-Lambert	Tier 2	A concentration phenomenon measured optically.
Vinegar + Baking Soda CO_2	Tier 2	Fast nonliving gas phenomenon.
Yeast / Sugar CO_2	Tier 2	Biological gas phenomenon, preparing for energy conversion thinking.
Fair Fitness	Tier 3	Human work, recovery, fairness, and ethical judgment.

Important correction: Purple Power and CandleLab / PopCan Calorimeter are associated only with DYNAMETICS / Tier 3. Kitchen Lab may prepare the variables used by those activities, but it should not absorb the activities themselves.

C. Tier 1 - MKS Measurement Grammar

Purpose: teach the language of measurement before any kitchen, fuel, or human-performance system is interpreted.

Rank	Experiment	Central phenomenon	Why it belongs here
1	Itty Bitty Measurement	Measurement has limits and resolution.	Opens the sequence without technology; establishes measurement humility.
2	Water Standard Lab	Water links grams, milliliters, and cubic centimeters.	Creates the physical anchor for all later mass/volume reasoning.
3	DIY Graduated Cylinder	Instruments are built from geometry, not magic.	Students understand a measuring device before using one blindly.
4	Smart eScale Calibration and Trust Lab	Zero, tare, repeatability, span, drift.	The scale must be trusted before gas, calorimetry, and fuel labs.
5	Story of π / Circle Measurement	Derived quantities from direct measurements.	Connects geometry to the real measured world.
6	Seconds Pendulum / Water Clock	Time, rate, and accumulation become visible.	Introduces changing processes rather than static measurements.
7	Logging Thermometer	Temperature becomes a changing measured variable.	Prepares for mixing, cooling, heating, and fermentation.
8	Tape-less Rule / Distance Sensor	Instrumented distance and non-contact measurement.	Useful extension of space measurement; supports later motion work.
9	UMA Dashboard / Data Logger	Measurements become records, graphs, and stored evidence.	Technology enters as witness and recorder, not as the star.

Tier 1 logic: Matter comes first, then time and rate, then temperature and digital instrumentation.

D. Tier 2 - Kitchen Lab

Purpose: use ordinary materials to study heat, concentration, gas production, and biological change.

Rank	Experiment	Central phenomenon	Why it belongs here
10	Predictive Mixing	Hot and cold water produce a measurable final equilibrium.	Simplest kitchen phenomenon: visible, concrete, and predictable.
11	Microwave / Hot Plate Water Heating	Water gains heat over time from a power source.	Adds energy input and rate to temperature change.
12	Newton's Cooling	Temperature change unfolds as a time-series curve.	Students move from point measurements to modeled change over time.
13	TeaTime / Beer-Lambert Lab	Concentration becomes visible through color and light.	Flagship concentration phenomenon; richly measurable and memorable.
14	Dilution / Calibration Lab	Known concentrations create a standard curve.	Makes TeaTime quantitative and predictive.
15	Vinegar + Baking Soda CO_2 Lab	A fast nonliving chemical reaction generates gas.	Best first gas lab: obvious, quick, and easy to reason about.
16	Yeast / Sugar CO_2 Generator Capture Lab	A slower biological process consumes sugar and produces gas.	Builds on gas logic but adds metabolism, time, and rate.
17	Temperature-Controlled Fermentation Extension	Heat history affects biological gas production.	Predictive mixing and Newton cooling now explain fermentation rate.

Tier 2 logic: Heat transfer comes first, concentration next, then fast gas and slow biological gas.

E. Tier 3 - DYNAMETICS / Bio-DYNAMETICS / Fair Fitness

Purpose: turn measurement into judgment: useful, efficient, affordable, humane, fair, and enough.

Rank	Experiment	Central phenomenon	Why it belongs here
18	CandleLab / PopCan Calorimeter	Fuel mass disappears while water gains useful heat.	First full DYNAMETICS synthesis: mass, heat, time, power, efficiency.
19	Appliance Efficiency Dashboard	Household energy use is judged by usefulness and loss.	Extends calorimetry logic into everyday decision-making.
20	Human Candle Comparison	Resting human metabolism can be compared to a slow candle.	Bridges machine/combustion thinking to the living body.
21	Purple Power DIY Gatorade Build	Human fuel can be designed and measured.	Purple Power belongs only here: DYNAMETICS / Bio-fuel.
22	Purple Power Energy / Cost per Serving	Fuel is judged for concentration, practicality, and affordability.	The drink becomes a useful-or-not-useful system.
23	Hydration and Fuel Use Discussion	Fuel and hydration are interpreted in human terms.	Converts recipe data into physiological meaning.
24	Purple Power -> Yeast CO_2 Demonstration	A DYNAMETICS fuel can be shown driving biological conversion.	Optional bridge demonstration; keep Purple Power conceptually in Tier 3.
25	Original Fair Fitness Step Test	Human work, cadence, power, and recovery are measured.	Human performance becomes measurable without humiliation.
26	Fair Fitness Testing vs PACER	Measurement should inform rather than shame.	Ethical interpretation only lands after real quantitative experience.
27	SQM+ / WalkWise Extension	Clinical-style gait and motion measurement.	Capstone: highest maturity of human measurement in the sequence.

Tier 3 logic: CandleLab synthesizes fuel, heat, and efficiency; Purple Power becomes measured human fuel; Fair Fitness measures work and recovery humanely.

F. Compact 12-Lab Flagship Sequence

For a compact course, these 12 labs preserve the whole conceptual spine.

Seq	Flagship lab
1	Itty Bitty Measurement
2	Water Standard + eScale Calibration
3	Seconds Pendulum / Water Clock
4	Logging Thermometer + Dashboard
5	Predictive Mixing
6	Newton's Cooling
7	TeaTime / Beer-Lambert
8	Vinegar + Baking Soda CO_2
9	Yeast / Sugar CO_2 Generator
10	CandleLab / PopCan Calorimeter
11	Purple Power DIY Gatorade / Bio-Fuel
12	Fair Fitness Step Test vs PACER

Compact course rule: Keep the memory anchors: MKS foundation, a temperature/time system, a concentration lab, a fast gas lab, a slow gas lab, a DYNAMETICS synthesis, and a humane human-performance measure.

G. Final Rationale

The curriculum should be remembered as a **story told with instruments**. Students first learn that measurement has limits and rules. Then they use those measurements to watch ordinary materials behave in revealing ways: water mixes, water cools, tea darkens, gas appears, and living systems consume sugar. Only after those phenomena are familiar does the curriculum advance into DYNAMETICS, where fuel is judged for usefulness, heat is judged for efficiency, and human performance is judged for fairness rather than humiliation.

This phenomena-first arrangement is stronger than a technology-first arrangement because it keeps the meaning of each lab in the foreground. The microcontroller, dashboard, and sensor suite remain important, but they are no longer the center of attention. They become witnesses and servants of the experimental question.

One-line master sequence: measure matter -> measure heat -> measure concentration -> generate gas -> judge useful energy -> measure fair work.